

Lab 1 - Rhythmiq

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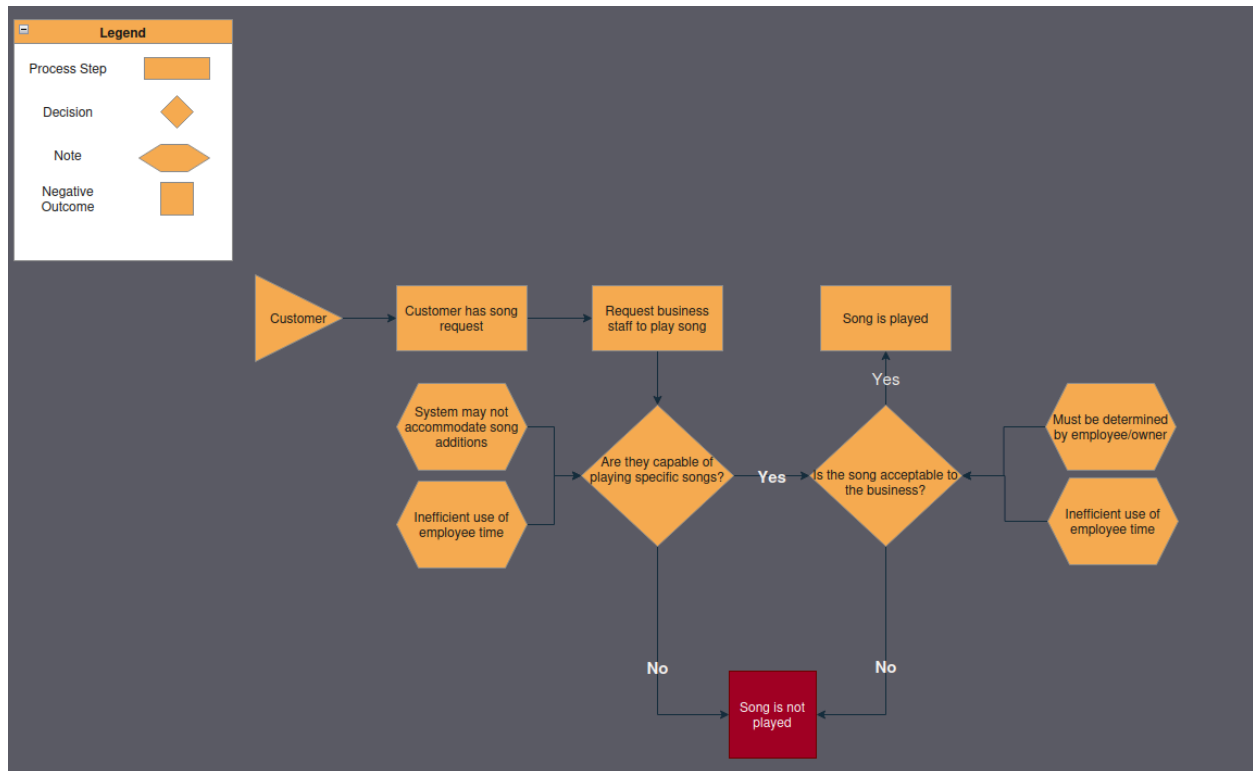
1. Introduction

At small cafes, coffee shops, local restaurants, and other places in the hospitality industry, the atmosphere of a venue directly impacts customer satisfaction and dwell time. Over one million independent food and beverage venues in the United States lack affordable, customer-interactive music management tools. Small businesses cannot justify spending large amounts of money on proprietary jukebox solutions.

Some solutions to this problem exist, but they all come with limitations. TouchTunes is a jukebox service, but it requires hardware, there is a high upfront cost, and it is designed for large bars and restaurants, being inaccessible to small businesses [TouchTunes, n.d.]. Rockbot is another similar service, recommends dedicated hardware, has a medium cost; targets medium to large businesses, and is limited in its mobile and web interface [Rockbot, n.d.]. Outside of dedicated services, customers can make manual staff requests, where customers verbally ask staff to change songs, but this is inefficient use of employee time during peak hours and is not consistent. Businesses may also use generic personal streaming (e.g., Spotify), but this is illegal for commercial use without a commercial license and it has no customer interaction feature.

There is a market gap for a music management solution that is affordable, real-time, staff-free, hardware-free, customer-interactive, and legally compliant for small local venues. Customers have no way to influence venue atmosphere digitally under these restrictions, resulting in owners missing a potential secondary revenue stream from music engagement. A solution to this problem must be cheap for a small business, usable without buying extra hardware, accessible for customers alone without any staff input, compliant with commercial music licensing laws, and staff must have control over final song selection.

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As a solution to address all of these issues, we are creating Rhythmiq. It is a web-based, hardware-agnostic, subscription jukebox platform for local spaces.

2. Rhythmiq Product Description

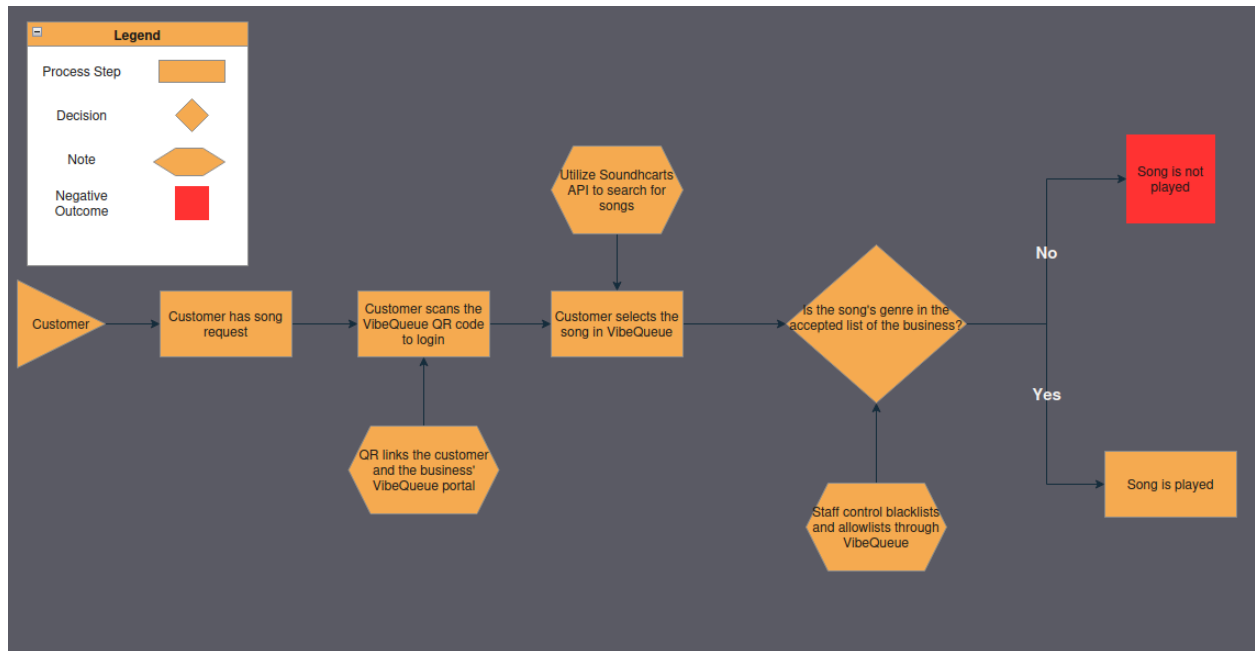
Rhythmiq is a software-only solution that replaces expensive proprietary hardware and manual staff processes with a seamless digital music ecosystem for small local spaces. It will give customers a frictionless way to influence the venue's music via their own smartphones, give owners and staff complete control over which songs can be requested, provide an optional secondary revenue stream with a pay-to-request model, ensure all music playback is legally licensed for commercial use, and be deployable with zero new hardware purchase.

2.1 Rhythmiq Product Description

Rhythmiq beats out its competitors because software only runs on devices the business already owns, it has a subscription-based low monthly cost vs. high upfront hardware

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investment, it has a customer facing QR interface vs. staff mediated requests, and it is legally compliant via SoundMachine (ASCAP/BMI/SESAC) vs. unlicensed personal streaming.



Customers scan a QR code on their table to access the venue's Rhythmiq portal in a mobile browser with no app download required, as the QR code links the customer's device to the business's specific portal. Customers search and browse songs using metadata provided by the Soundcharts API, and results are filtered in real time based on the venue's active vibe constraints. There is a proprietary matching algorithm built by the Rhythmiq team, which evaluates songs against venue's allowed genre, tempo, mood, and popularity settings, and rejects non-matching requests before they reach the queue. This is significant because it replaces the need for the Spotify API, which has strict commercial use restrictions, with a legally safe, in-house solution. This system allows venues to maintain brand atmosphere even with customer input.

2.2 Major Components

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From the business control center, there is a core management interface, the owner/staff-facing dashboard. It allows you to set genre allowlists and blocklists (e.g., “acoustic only,” “no hip hop”), block explicit content automatically, manage the live song queue (skip, reorder, remove), manage user sessions (ban abusive users), and view analytics (most requested songs, peak request times).

There are integrated content filters applied system-wide, and owners can configure tempo range, mood tags, and explicit content rules once, which the system enforces automatically on every request.

There is an optional monetization feature where customers pay a small fee (e.g., \$0.99–\$1.99) to boost a song to the top of the queue, with automated payment processing. The business keeps a majority of the money, while Rhythmiq takes a 5% platform cut to sustain development. No competitor offers customer-facing paid interaction at this price point for small businesses.

All audio playback is routed through SoundMachine, which holds commercial music licenses (ASCAP, BMI, SESAC) and ensures Rhythmiq is fully legal for business use.

Rhythmiq runs on any internet-connected device the business already owns, such as staff phones, existing tablets, or a dedicated laptop. No proprietary or dedicated hardware is required.

The venue speaker system is connected to the device running SoundMachine.

The web app UI is a customer-facing interface accessed via a QR code. It is built as a responsive web app and has a search interface connected to the Soundcharts API. The “Vibe Engine” is a custom-built song selection and validation logic. It receives a customer song request, queries Soundcharts for metadata, checks against Venue Database constraints, and finally returns a validated/rejected result. The Soundcharts API is a third-party music analytics

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service that provides commercially safe metadata (genre, tempo, mood, popularity, explicit flag) for song lookup and vibe matching. SoundMachine is a third-party licensed commercial streaming service that handles actual audio playback in the venue and ensures ASCAP/BMI/SESAC compliance. The local server handles audio signal routing, playback status feedback, and communication between Vibe Engine and SoundMachine. The venue database stores venue-specific vibe settings (genre rules, explicit flags, tempo limits), the song queue state, and transaction/revenue data. The Business Control Center Dashboard is an owner/staff interface for real-time queue management, constraint configuration, and analytics viewing.

Rhythmiq uses the LAMP tech stack (Linux, Apache, MySQL, PHP), Soundcharts API music metadata, SoundMachine API-licensed audio playback, a QR code generation library with venue-specific portal linking), and a web-based front end with mobile-first responsive design. The initial prototype explored Spotify API, but the team pivoted due to strict Spotify commercial use licensing restrictions, and the current architecture uses Soundcharts and SoundMachine to eliminate legal risk.

3. Identification of Case Study

Brewpoint Café is a single-location independent coffee shop, run by the owner (Jamie) and two barista staff. It seats about 30 people, primarily serving morning commuters and remote workers from 7 AM to 2 PM.

Baristas are interrupted 10–15 times per morning rush by verbal song requests. The owner loses control of the café's intended "acoustic/lo-fi" vibe when staff accommodate requests, and no revenue is generated from music engagement.

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In this scenario, we are testing QR code access (on each table card), song browsing & requests (Soundcharts-powered search), the Business Control Center (genre restrictions: acoustic, indie, lo-fi only; explicit content blocked; max tempo 120 BPM), dynamic vibe moderation (auto-enforcement of genre and tempo rules), and pay-to-request (\$0.99 boost option during morning rush).

In this scenario, a customer “Alex” sits down, notices a QR card on the table, and scans it with their phone. The Rhythmiq portal opens in the browser, with no download needed, and Alex sees the current queue and a search bar. Alex searches “Golden Hour” by JVKE, which the Vibe Engine checks. The genre is pop/indie (pass), is not explicit (pass), and has a tempo of 98 BPM (under 120 BPM limit, pass). The song is added to the queue. Alex optionally pays \$0.99 to boost the song to the top of the queue, and the payment is processed automatically. The staff receives zero interruption and the song plays through SoundMachine on café speakers. The owner Jamie logs into the control center on the counter tablet during a slow period, then reviews the queue, sees request analytics, and adjusts the genre allowlist for the afternoon shift.

[Figure 3: Case Study scenario flow Alex scans QR > searches song > Vibe Engine validates > song queued > SoundMachine plays > staff uninterrupted]

In the future, we would like to expand by scaling to small cafe chains (5–10 locations) using a centralized Control Center with per-location vibe profiles. We would also like to expand to adjacent markets, such as college dining halls, hotel lobbies, waiting rooms, fitness studios, and barbershops. We would like to also support a potential B2B SaaS licensing model for venue management companies. We may also add loyalty/gamification features, like customers earning “vibe points” for requests that stay in the queue.

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4. Glossary

5. References

Rockbot. (n.d.). *Rockbot business music platform*. <https://rockbot.com/go>

Soundcharts. (n.d.). *Music analytics platform*. <https://soundcharts.com>

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